1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Do breast tumors with a high stromal-to-tumor ratio and harboring TP53 mutations with reduced CDH1 expression undergo EMT-driven invasion that mechanistically links loss of cell–cell adhesion to pro-invasive stromal remodeling, leading to accelerated local progression? If this is true, tumors with the TP53 mutation and low CDH1 in a high-TSR context will have shorter progression-free intervals in TCGA-BRCA compared with tumors lacking one or more of these features, identifying a biologically coherent subset at high risk of early progression.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, CDH1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 is frequently mutated in breast, both consistent with a conserved TP53–CDH1 axis modulating invasiveness; stromal remodeling (reflected by stromal-to-tumor ratio) is a plausible imaging correlate of EMT-driven desmoplasia.

CDH1: CDH1 axis modulating invasiveness; stromal remodeling (reflected by stromal-to-tumor ratio) is a plausible imaging correlate of EMT-driven desmoplasia.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Klebsiella	Supporting modifier from knowledge graph
Secondary	EMT	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Analyze TCGA-BRCA and independent datasets to quantify TSR from histology, determine TP53 mutation status, and measure CDH1 expression;

Prediction 2: assess EMT activity via gene signatures and key protein markers;

Prediction 3: test for a three-way interaction (TSR x TP53 status x CDH1 expression) on progression-free interval using Cox models, with replication in external cohorts;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- Klebsiella: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EMT: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.